

Revision Notes for Class 8 Science Chapter 3 Health The Ultimate Treasure

Health is one of the most valuable treasures in life. It is not just about avoiding sickness but also about physical fitness, mental strength, and positive relationships. The World Health Organization (WHO) defines health as a state of complete physical, mental, and social well-being—not merely the absence of disease. Feeling energetic, thinking clearly, and having good relations with others are as important as not having any illness in the body.

Importance of Mental, Physical, and Social Health

A healthy lifestyle means caring for your body and your mind. Eating fresh, balanced food, getting enough sleep, exercising, and sharing time with friends and family all help maintain health. For example, regular yoga and meditation contribute to mental well-being, while proper hygiene keeps diseases away. India's heritage, like Ayurveda, emphasizes the balance of body, mind, and surroundings for overall wellness.

Good Habits for a Healthy Lifestyle

Adopting healthy habits is key. Some essential points include:

- Maintaining personal hygiene and cleanliness.
- Following a balanced diet rich in fruits, vegetables, and whole grains.
- Exercising regularly and staying active.
- Sleeping on time and getting enough rest to recover each day.
- Saying no to harmful things like tobacco, alcohol, and drugs.
- Managing stress through relaxation or meditation.

Unhealthy habits like too much screen time, eating junk food, skipping meals (especially breakfast), and sleeping late can harm health. It is equally important to reduce loneliness by spending time with loved ones and taking part in social activities.

Environmental Factors in Health

Where we live affects our health. Clean surroundings, safe drinking water, and fresh air keep diseases away. Polluted areas lead to cough, asthma, and infections. Air pollution, especially in cities, is a growing concern. Poor hygiene, irregular handwashing, and dirty water can increase chances of falling ill. Communities are healthier when everyone helps to keep homes, schools, and public areas clean.

How Do We Know If We Are Unwell?

The body shows signals when something is wrong. Symptoms are what a person feels, like pain or tiredness. Signs are visible or measurable, such as fever, rashes, or swelling. Knowing the difference helps doctors find the correct cause and suggest the right treatment. Common warning signs include dizziness, headaches, and abnormal changes in appetite or weight.

Types and Causes of Diseases

There are two main types of diseases:

- **Communicable diseases:** Caused by germs like bacteria, viruses, fungi, and worms, and they spread from person to person. Examples—chickenpox, flu, dengue, cholera, typhoid.
- **Non-communicable diseases:** Do not spread from one person to another. They result from lifestyle factors (like unhealthy food, lack of exercise, or pollution) and include diabetes, cancer, and asthma.

In India, non-communicable diseases are rising rapidly, mainly due to poor diet, less exercise, and stressful living. Both types of diseases require different methods for prevention and control.

How Do Diseases Spread?

Communicable diseases can spread in many ways:

- Through air—sneezing and coughing (like flu, measles, tuberculosis).
- By contaminated food and water (like cholera, hepatitis A, typhoid).
- Via insects called vectors, such as mosquitoes spreading malaria and dengue.
- Direct contact with an infected person or sharing their items.

Prevention includes covering the mouth and nose while sneezing, washing hands often, boiling drinking water, using mosquito nets, and maintaining hygiene.

Table: Some Common Communicable Diseases

Disease	Cause	Source	Key Symptoms	Prevention
Common cold	Virus	Air, contact	Cough, runny nose, fever	Handwashing, covering mouth
Cholera	Bacteria	Water/food	Diarrhoea, dehydration	Clean water, hygiene
Malaria	Protozoa	Mosquito bite	Fever, chills	Mosquito nets

Non-Communicable Diseases and Their Prevention

Diseases like obesity, diabetes, and hypertension are not caused by germs and do not spread person-to-person. They develop slowly and can last for years (chronic). Their major causes are unhealthy diets, little physical activity, stress, and sometimes genetics. For example, diabetes is very common in India now because of increased fast food, reduced exercise, and more screen time. Preventing non-communicable diseases is about eating right, exercising regularly, and managing stress.

Body's Defenses: Immunity and Vaccines

Immunity is the body's power to fight diseases. Some people do not fall ill easily because their immunity is strong. Vaccines are special medicines (given as drops or injections) that help the immune system recognize and fight germs. For example, vaccines protect against polio, measles, and tetanus by training the body to recognize these infections in advance. India's tradition of variolation against smallpox is an early example of trying to prevent disease before it spreads.

Modern vaccines are safe and important for everyone. When a large proportion of people are vaccinated, the spread of diseases slows down. Vaccines are preventive but not curative—they reduce risk before infection happens but cannot treat a disease once someone is already sick.

Treatment and Responsible Use of Antibiotics

Most diseases can be treated by consulting a doctor. For bacterial infections, doctors prescribe antibiotics that kill the bacteria. However, these medicines do not work on viral diseases like cold or flu. Overusing antibiotics leads to antibiotic resistance, where bacteria stop responding to medicines. This can make infections harder to treat in the future. Always take antibiotics as advised by the doctor, in the right dose and for the full duration.

Traditional Indian systems like Ayurveda and Unani also use natural herbs and focus on a balanced lifestyle. These systems can help in building health, but are not always suitable for all illnesses, especially rapidly progressing or serious diseases.

How to Prevent and Control Diseases

The best defense is to prevent sickness before it starts. This means:

- Practising good sanitation, using toilets, and maintaining a clean environment to prevent outbreaks like diarrhoea and cholera.
- Good hygiene, like frequent handwashing, covers most infectious diseases.
- Safe water and well-cooked food protect against many illnesses.

- Getting timely vaccinations keeps serious infections away.
- Healthy habits—balanced diet, outdoor play, rest, and limited junk food—cut risk of non-communicable diseases.
- Limiting the use of antibiotics unless necessary to prevent resistance.

Key Takeaways

- True health involves physical strength, a cheerful mind, and supportive social connections.
- Diseases can be short-term or chronic, infectious (from germs) or non-infectious (from lifestyle/environment).
- Vaccines, hygiene, and better living conditions protect against many diseases.
- Mental and emotional health are as important as a healthy body for happiness and success.

Taking small but regular steps—like eating on time, drinking clean water, and talking to loved ones—builds good health that lasts a lifetime. By understanding both prevention and treatment, everyone can make smarter choices for a healthy future.